

Physics 20A: Astronomy (Winter 2026) Final Projects

Due date: Friday March 13th, 11:59 pm CA time.

This final project replaces the final exam scheduled for this class by the Registrar.

The final projects involve any creative activity on any topic related to astronomy and space exploration, either already discussed in class or not covered in class. The possibilities are numerous and activities can range from a web site, an entry to Wikipedia, to writing and composing a song, writing a play, writing a movie synopsis, making a video that you can upload to YouTube, among many other possibilities that I haven't even considered.

The only requirement is your project must be original and must involve about 20 hours of work in total per person, spread over 3-4 weeks (ie 6-8 hours a week till the last week of classes). Projects may be completed individually or as groups of not more than 3 students, but the complexity of the project must then be proportional to the number of you involved in that project. If group projects, you must write and attach a short statement on what the individual contributions were to that project. TA will assign the same score to all students in a group.

The only thing that will not be accepted as a final project are typed-up descriptive papers that simply summarize what we already know about a planet or planets, stars etc. These are typically the type of papers that you may have completed in high school and are easy to complete with Google. Instead of simply typing up a research paper on existing knowledge, I will accept projects that have original ideas and creativity, and show your own work clearly.

The final projects are due on Friday, March 14th. There are no exceptions or delays to this deadline. If you cannot meet this deadline, I will assign an incomplete grade unless you email and say a grade based on the work done without a final project is acceptable.

Depending on your exact project you will submit for example:

1. If a website, the URL and an electronic copy of the website saved as a PDF to Canvas.
2. If some creative activity, a copy of your work – could be photos that are saved as a PDF to Canvas.
3. A video of the project completed (like a book, make a video of showing/describing the book) and upload to YouTube or direct to Canvas.

Final project submissions should be roughly twice the length of Assignments I and II. If written, it should be about 8-10 pages of material. If video for example I expect it will be about 6-8 minutes. Again, there are no exact length requirements. Again, it should be roughly twice the work of Assignment I and II.

Here are some ideas for the final project, **but this list is not complete. There is no requirement or expectation your project will be from this list.**

1. A short movie synopsis about human exploration of the Solar system or a planet or a moon of the solar system, without plagiarizing movies already made. (In this case you

will submit the typed-up synopsis/script. A movie synopsis is not simply writing a story, you must also introduce the characters, provide background information, and a summary of the story. A completed movie script will involve the full dialogue between characters as they would read it on set during the production. This is not necessary since I would not expect you to write such a script in 4 weeks).

Alternatively, you can create a video to describe the movie synopsis.

2. A website summarizing Mars rovers so far and how you think future Mars exploration should look like. (in this case you will submit the URL and also a copy of the website saved as a PDF file).
3. A fictional short story on finding life on another planet. (In this case you will submit the story typed up).
4. A non-fictional play/act describing a future activity at NASA Mission Control Office, trying to recover a mission to Venus' top atmosphere that went to look for evidence for bacteria responsible for phosphene signature. (In this case you will submit the act written up, this will be similar to movie script).
5. A song on cosmological models at the time of Galileo, for example, or anything in astronomy/cosmology/space. (In this case you will submit the song and the musical score for instruments involved. you may also compose a non-vocal piece).
6. An idea for a new TV show centered on astronomy/cosmology/space. This would be like #1 on a movie synopsis. If it is really good, I will put you in touch with a production company that makes series for Netflix and other streaming services.
7. A new children's book on astronomy/cosmology/space – you can design the book and submit a video showing the book or photos of the book saved as a PDF to Canvas. If it is really good you can self-publish on Amazon and I can put you in touch with an agent who will help you guide that process.

You will retain copyrights to all projects that you submit. Once again, the projects must be original, your or your group's own work. This is the only thing I emphasize. There are many other possibilities than the ones I list. These are just suggestions, please be creative and design a project of your own.

Submission instructions: submission is via Canvas, but for some reason Canvas may not accept certain files. In such a case save on a Google drive and upload a PDF with link.

If the submission is a group project, please have only one of member of the group submit the project. We do not need multiple submissions of the same project. Please include a clear cover sheet that lists all names of group members and a statement that all members contributed equally. If other group members prefer they are free to also submit the same cover sheet only and identify the name of the student that is submitting the full project.